

HUJI Research Monitor

Weekly Digest · Week 29 · July 13, 2026

EXECUTIVE SUMMARY

Hebrew University researchers are advancing a diverse portfolio of innovations poised for significant commercial impact. This digest highlights breakthroughs ranging from novel medical devices enhancing cardiovascular procedures and precision diagnostics for complex diseases like fibromyalgia and Crohn's, to sustainable technologies transforming plastic waste into valuable products and dramatically reducing cultivated meat production costs. We also feature cutting-edge AI platforms for drug discovery and mental health monitoring, alongside fundamental material science advancements in quantum dots. These innovations underscore our commitment to addressing critical global challenges and market needs with high-potential solutions.

#1

First-in-Human Mechanical Leaflet Modification of Native Aortic Stenosis During TAVR.

36/50

Main Researcher: Anna Turyan

[Medical Device] [Clinical]

"Novel mechanical device enhances TAVR safety by preventing coronary obstruction, unlocking a new market segment."

WHY NOW

The rapidly expanding TAVR market urgently needs solutions to address critical procedural complications like coronary obstruction, ensuring wider patient eligibility and safety.

COMMERCIAL ANGLE

The expansion of mechanical leaflet modification tools from bioprosthetic valves to native aortic stenosis represents a significant new market segment for transcatheter valve intervention technologies.

<https://pubmed.ncbi.nlm.nih.gov/42340176/>

#2

Growth factors-infused cellulose scaffolds support cost-efficient proliferation and differentiation of bovine stem cells for cultivated meat.

35/50

Main Researcher: Sharon Schlesinger

[FoodTech] [AgriTech] [Materials] [Synthetic Biology]

"Cellulose scaffolds slash cultivated meat production costs by reducing expensive growth factors tenfold."

WHY NOW

High growth factor costs are the primary barrier to cultivated meat's commercial viability; this innovation directly enables scalable, affordable production.

COMMERCIAL ANGLE

This technology addresses the single largest cost barrier in cultivated meat production by significantly reducing the consumption of expensive growth factors through localized scaffold delivery.

<https://europepmc.org/article/MED/42182043>

#3

Adaptive Laboratory Evolution Unlocks Membrane Permeability as a Key Limitation in Long-Chain Alcohol Metabolism by *Pseudomonas putida* KT2440.

34/50

Main Researcher: Lianet Noda-García

[Materials] [Synthetic Biology] [Clean Energy]

"Engineered bacteria transform plastic waste into valuable chemicals, driving a circular bio-economy."

WHY NOW

Global mandates and consumer demand for plastic waste solutions are immense, creating a massive market for technologies that valorize waste into sustainable feedstocks.

COMMERCIAL ANGLE

This technology enables the development of circular bio-economies by transforming plastic waste degradation products into valuable microbial feedstocks for high-value chemical production.

<https://pubmed.ncbi.nlm.nih.gov/42338148/>

#4

Allele-specific antisense oligonucleotide treatment rescues

34/50

Main Researcher: Tamar Harel

[Drug Discovery] [Genomics] [Clinical]

"Precision antisense oligonucleotides selectively target disease-causing mutations for safer genetic therapies."

WHY NOW

Advances in genomics demand highly specific therapies for rare genetic disorders, reducing off-target effects and opening new avenues for personalized medicine.

COMMERCIAL ANGLE

The ability to precision-target single-nucleotide mutations opens a high-value therapeutic pathway for rare genetic disorders and personalized genomic medicine.

<https://pubmed.ncbi.nlm.nih.gov/42239176/>

#5

Progress in structure prediction and design of adaptive immune receptors.

30/50

Main Researcher: Dina Schneidman-Duhovny

[Drug Discovery] [Immunology] [Software/AI] [Synthetic Biology]

"AI-driven platform designs custom immune receptors, accelerating personalized cancer immunotherapy."

WHY NOW

The boom in immunotherapy demands faster, more precise methods for designing effective treatments, a critical need that AI can now address.

COMMERCIAL ANGLE

Developing AI-driven platforms for the precision design of synthetic antibodies and T-cell receptors for personalized cancer immunotherapies.

<https://pubmed.ncbi.nlm.nih.gov/42430862/>

#6

Plasmon-Induced Hot-State Multiexciton Emission from Quantum Dots Coupled to Metallic Nanocavities.

25/50

Main Researcher: Uri Banin

[Materials] [Imaging] [Quantum]

"Breakthrough quantum dot technology achieves ultra-efficient light emission for next-gen photonics and sensors."

WHY NOW

Persistent demand for brighter, more efficient light sources and hypersensitive optical sensors across industries, from displays to deep-tissue medical imaging.

COMMERCIAL ANGLE

Development of ultra-efficient, high-power photonic devices, next-generation LEDs, and hyper-sensitive optical sensors for deep-tissue imaging.

<https://europepmc.org/article/MED/42189911>

#7

Nitric oxide inhibition ameliorates cortical proteomic changes in the Cntnap2^{-/-} and Shank3^{Δ4-22} mouse models of autism spectrum disorder.

25/50

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Proteomics] [Genomics]

"Identified nitric oxide pathway offers a novel small-molecule target for genetically-linked autism spectrum disorders."

WHY NOW

There's a significant unmet need for effective, targeted small-molecule treatments for autism, with current options largely symptomatic.

COMMERCIAL ANGLE

The identification of specific nitric oxide signaling pathways provides a validated pharmacological target for developing first-in-class small molecule treatments for genetically-linked autism spectrum disorders.

<https://europepmc.org/article/MED/42010670>

#8

Haptoglobin, hemolysis, and renal vulnerability after pulsed field ablation: The hemoglobin-binding reserve.

23/50

Main Researcher: Rabea Asleh

[Medical Device] [Diagnostics] [Clinical]

"Diagnostic screening tool identifies patients at renal risk from new cardiac ablation procedures."

WHY NOW

The rapid adoption of Pulsed Field Ablation requires companion diagnostics to improve patient safety and mitigate known risks, ensuring broader clinical use.

COMMERCIAL ANGLE

Development of a pre-operative diagnostic screening tool or companion diagnostic to identify patients with low haptoglobin reserves who require protective measures before PFA.

<https://pubmed.ncbi.nlm.nih.gov/42235779/>

#9

Serum from Fibromyalgia Patients Activates Satellite Glial Cells in Mouse Peripheral Ganglia.

23/50

Main Researcher: Suhail Amar

[Diagnostics] [Drug Discovery] [Neuroscience] [Immunology]

"Blood-based biomarker panel targets circulating factors for objective diagnosis and therapy of fibromyalgia."

WHY NOW

Fibromyalgia lacks objective diagnostics and effective treatments; a blood-based approach offers a major breakthrough for millions of sufferers.

COMMERCIAL ANGLE

Development of a blood-based diagnostic biomarker panel or targeted immunotherapies to neutralize the specific serum factors causing glial activation.

<https://pubmed.ncbi.nlm.nih.gov/42274567/>

#10

Strain-level microbial signatures and inferred functional alterations in infants with food protein-induced allergic proctocolitis.

23/50

Main Researcher: Moran Yassour

[Diagnostics] [Genomics] [Immunology] [Software/AI]

"Microbial signatures enable precision diagnostics and therapies to prevent lifelong infant food allergies."

WHY NOW

Growing public health concern over food allergies drives demand for early, non-invasive diagnostic tools and preventative microbiome-based interventions.

COMMERCIAL ANGLE

The identified strain-level microbial signatures provide a foundation for developing precision diagnostic kits and microbiome-based therapeutic interventions to prevent lifelong food allergies.

<https://pubmed.ncbi.nlm.nih.gov/42010622/>

#11

Temporal patterns of smartphone-mediated digital engagement and mental health symptomatology.

23/50

Main Researcher: Amit Birenboim

[Software/AI] [Diagnostics] [Neuroscience] [Clinical]

"Smartphone digital phenotyping offers real-time mental health monitoring and predictive intervention."

WHY NOW

The global mental health crisis and widespread smartphone use create an urgent market for accessible, AI-driven tools for early detection and personalized support.

COMMERCIAL ANGLE

Developing digital phenotyping software for real-time mental health monitoring and predictive intervention via smartphone sensor data.

<https://pubmed.ncbi.nlm.nih.gov/42364396/>

#12

22/50

Multi-omics-based machine learning model predicts response and guides treatment in Crohn's disease: a case study in nutritional therapy-response to editorial.

Main Researcher: Dan Turner

[Diagnostics] [Software/AI] [Genomics] [Clinical]

"Multi-omics AI model predicts Crohn's nutritional therapy response for personalized IBD management."

WHY NOW

Chronic IBD requires personalized care; this AI-driven diagnostic guides precision nutrition, offering a less invasive and highly effective treatment path.

COMMERCIAL ANGLE

Development of a companion diagnostic software platform for precision nutrition and personalized medicine in Inflammatory Bowel Disease (IBD) management.

<https://europepmc.org/article/MED/42133586>